

MR Boundary Theory — Zero as Boundary Operator (Roles, μ , and Spectral Tests)

Status: atlas-ready formal component (v1.0)

Scope: resolves “Zero” ambiguity by role-decomposition; links algebraic boundary notions to MR semigroup dynamics and spectral falsifiability.

0. Executive summary

A single symbol “0” can play distinct mathematical roles depending on the operation $\circ$:

- **Unit (identity):** does nothing under composition.
- **Absorber (annihilator):** collapses outputs.
- **Projector/closure (boundary operator):** maps states/processes into a lawful subspace.
- **Categorical zeros:** zero morphisms / zero objects (context-dependent).

This component replaces “Zero” with a **role schema** (left/right units and boundaries) governed by a **monotone** μ (“non-creativity” / lawfulness deviation). In MR, choosing $\circ$ as **process composition** and μ as **illegal-mass / deviation from lawful fixed space** produces **spectral predictions** (gap vs. no-gap) that are testable via prime truncations.

1. Motivation and the core disambiguation

The statement “zero marks the boundary where an operation closes without remainder” becomes rigorous once “zero” is treated as a **family of roles**, not a single object.

Examples (same symbol, different roles):

- $\emptyset$ is a **unit** for union: $A \cup \emptyset = A$, but an **absorber** for intersection: $A \cap \emptyset = \emptyset$.
 - In categories, **identity morphisms** exist for every object; **zero morphisms** exist only in special categories and are not identities.
 - In physics, “vacuum” is typically a **fixed point / ground state** (a boundary *sector*), not automatically an identity element.
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2. Primitive boundary roles (symmetric schema)

Work in a **typed partial algebra** $(X, \circ)$ where $x \circ y$ may be undefined unless types match.

2.1 Monotone

Let $\mu : X \rightarrow (M, \preceq)$ be a monotone into an ordered commutative monoid (e.g., $M = \mathbb{R}_{\geq 0}$, $\oplus = +$, $\preceq = \leq$).

Interpretations:

- **Cost** (effort/energy/information cost)
- **Lawfulness deviation** (distance from lawful subspace)
- **Resource / creativity budget**

2.2 Roles

For composable pairs (where expressions are defined):

- **Right-unit:** $x \circ b = x$
- **Left-unit:** $b \circ x = x$
- **Right-boundary:** $\mu(x \circ b) \preceq \mu(x)$
- **Left-boundary:** $\mu(b \circ x) \preceq \mu(x)$

Full unit: both unit equalities hold.

Full boundary: both boundary inequalities hold.

2.3 Emergent special cases

- **Absorber / annihilator:** $b \circ x = b$ and/or $x \circ b = b$
- **Projector/closure boundary:** $b \circ x = \pi(x)$ with $\pi^2 = \pi$ and $\mu(\pi(x)) \preceq \mu(x)$

To avoid μ becoming a mere label, impose at least one compatibility law.

Option A (cost subadditivity)

$$\mu(x \circ y) \preceq \mu(x) \oplus \mu(y).$$

Option B (lawfulness Lyapunov / contractive)

A left-action contractivity form:

$$\mu(x \circ y) \preceq \mu(y) \quad \text{for admissible left actions } x.$$

(or symmetrically on the right).

Interpretation: “non-creative” actions do not increase illegal mass / deviation.

4. MR instantiation: dynamics as process composition

4.1 Choose the operation

Let $\circ$ be **sequential composition** of MR processes. For a semigroup $(\Phi_t)_{t \geq 0}$:

$$\Phi_t \circ \Phi_s := \Phi_{t+s}, \text{quad } \Phi_0 = I.$$

4.2 Generator and lawful sector

Let G be the generator (e.g., $\Phi_t = e^{tG}$). In the MR positivity-certified dissipative regime, take $G = -U$ where $U \succeq 0$ and $U = A + B + E$ is the universal operator.

Define the lawful fixed space:

$$H_{\text{law}} := \text{Fix}(\Phi) = \{\psi : \Phi_t \psi = \psi \text{ for all } t \geq 0\}.$$

Let Π_{law} be the orthogonal projector onto H_{law} .

4.3 Canonical MR monotone (illegal mass)

Define

$$\mu(\psi) := \|(I - \Pi_{\text{law}})\psi\|^2.$$

MR also uses an explicit residual/projection decomposition in its recursion constitution, where “illegal residual” quantifies deviation from the prime-lawful projection; this supplies an implementation-ready μ variant.

5. Boundary Classification Theorem (atlas statement)

Theorem (Boundary Classification — MR process form)

Let $(\Phi_t)_{t \geq 0}$ be a strongly continuous **contraction semigroup** on a Hilbert space H with generator $G = -U$, where U is self-adjoint and $U \succeq 0$. Let $H_{\text{law}} = \text{Fix}(\Phi)$, Π_{law} its orthogonal projector, and $\mu(\psi) = \|(I - \Pi_{\text{law}})\psi\|^2$.

1. **Unit uniqueness:** $I = \Phi_0$ is the unique full unit in $\{\Phi_t : t \geq 0\}$.

2. **Boundary monotonicity:** for all $t \geq 0$ and $\psi \in H$,

$$\mu(\Phi_t \psi) \leq \mu(\psi).$$

Moreover, $\mu(\Phi_t \psi) = 0 \forall t$ iff $\psi \in H_{\text{law}}$.

1. Absorber characterization (projector absorbers): an operator $P \in B(H)$ is a (lawful) absorber iff

$$P^2 = P, \text{Ran}(P) \subseteq H_{\text{law}}.$$

Equivalently, $(I - \Pi_{\text{law}})P = 0$ and P is idempotent.

1. Spectral-boundary correspondence: define

$$\lambda_* := \inf \{ \text{Re}(\lambda) : \lambda \in \sigma(U) \}.$$

• **(Gap case)** If $\lambda_* > 0$, then for all ψ :

$$\mu(\Phi_t \psi) \leq e^{-2\lambda_* t} \mu(\psi).$$

Hence relaxation time $\tau \sim 1/\lambda_*$.

• **(No-gap case)** If $\lambda_* = 0$ (in particular if $0 \in \sigma_{\text{ess}}(U)$), then uniform exponential relaxation fails; there exist sequences of states with arbitrarily slow decay of $\mu(\Phi_t \psi)$.

Proof sketch (what to include in the atlas)

- (1) follows from standard semigroup unit uniqueness.
- (2) follows because Π_{law} projects onto fixed points, and contraction implies non-expansive action on the complement.
- (3) is elementary: idempotent maps with range inside the fixed sector are precisely closure/absorber maps into lawfulness.
- (4) is the standard spectral-gap $\Rightarrow$ exponential convergence estimate on $H_{\text{law}}^\perp$ for self-adjoint $U \succeq 0$, and the converse uses approximate eigenvectors near 0 in the gapless/essential-spectrum case.

6. Falsifiable predictions

6.1 Core spectral fork (the “one computation” test)

On finite prime truncations U_N :

- Compute eigenvalues near 0.
- Estimate $\lambda_{\min}(N)$ = smallest nonzero eigenvalue (in the dissipative regime).

Predicted asymptotics as $N \rightarrow \infty$:

- **Gap scenario:** $\lambda_{\min}(N) \rightarrow \lambda_* > 0$ and relaxation remains uniformly exponential.

- **Continuum scenario:** $\lambda_{\min}(N) \rightarrow 0$ and relaxation becomes arbitrarily slow.

6.2 Dynamical check

Simulate $\psi(t) = \Phi_t \psi(0)$ and measure $\mu(t) = \mu(\psi(t))$:

- Gap $\Rightarrow \log \mu(t)$ is asymptotically linear with slope $-2\lambda_*$.
- No-gap $\Rightarrow$ non-linear tails; no single global exponential fit.

7. Validation protocol (implementation-ready)

7.1 Minimal spectral test (pseudocode)

```
def validate_boundary_prediction(G):
    evals = spectrum(G) # near 0
    evals = sorted(evals, key=lambda z: abs(z))

    # assume evals[0] ~ 0
    gap = abs(evals[1])
    if gap > GAP_THRESHOLD:
        return {"regime": "ISOLATED_GAP", "lambda_min": gap}

    # density near 0 as crude continuum proxy
    zero_near = sum(1 for e in evals if abs(e) < EPSILON)
    if zero_near / len(evals) > DENSITY_THRESHOLD:
        return {"regime": "ESSENTIAL/CONTINUUM", "count": zero_near}

    return {"regime": "AMBIGUOUS", "lambda_min": gap}
```

7.2 Cross-truncation scaling (most diagnostic)

For increasing truncations N :

- compute $\lambda_{\min}(N)$
- compute fitted $\tau(N)$ from $\mu(t)$ decay
- test $\tau(N) \approx 1/\lambda_{\min}(N)$

7.3 Consistency checklist

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8. Suggested MR Atlas placement

MR Atlas → Formal Components → Boundary Theory

1. Mathematical framework

2. Primitive roles (left/right unit/boundary)

3. μ compatibility axioms

4. Boundary Classification Theorem

5. Physical instantiation

6. $\circ$ = process composition (semigroup)

7. μ = illegal mass / deviation from lawful sector

8. spectral predictions

9. Validation protocol

10. spectral test (gap vs essential)

11. dynamical simulation (relaxation fitting)

12. cross-truncation scaling test

9. Internal MR references (project files)

- **Meta_Relativity.pdf** — universal operator $U = A + B + E$, positivity certification, essential spectrum criterion.
 - **Meta-relativity Atlas Of Frames.pdf** — MR atlas structure and operator framing.
 - **ΞConstitution.pdf** — prime projection Π_P , illegal residual $r_\perp$, and lawfulness metric construction.
 - **Meta-relativity Lawful Curvature Report.pdf** — fixed-point projector usage and lawful restriction geometry.
 - **Three Equals One-meta-relativity-appendix.pdf** — truncation workflow, certification-to-timescale link ($\tau \sim 1/\lambda_{\min}$).
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10. External references (standard math/physics foundations)

Category theory / algebraic structure

- S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer (Graduate Texts in Mathematics 5), 1998.

Semigroups and generators

- K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer (Graduate Texts in Mathematics 194), 2000.
- E. B. Davies, *One-Parameter Semigroups*, Academic Press (LMS Monographs), 1980.
- E. Hille and R. S. Phillips, *Functional Analysis and Semi-Groups*, American Mathematical Society, 1957. (Classic reference for Hille–Yosida.)

Functional analysis / spectral theory

- M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. IV: Analysis of Operators*, Academic Press / Elsevier, 1978.

Convergence and spectral gap intuition (useful for exposition)

- References on Poincaré inequalities / spectral gap and exponential convergence for reversible semigroups (standard in Markov semigroup and functional inequality literature).

Appendix A — Terminology alignment (“Zero” vs “Boundary”)

When writing MR-facing prose, avoid saying “the zero element” unless the operation is explicitly fixed.

Recommended language:

- **Unit process:** I or Φ_0
- **Lawful projector / closure:** Π_{law}
- **Absorber:** idempotent map into H_{law}
- **Boundary sector:** H_{law} (fixed space / kernel sector)

This preserves categorical hygiene while retaining the original intuition: boundary = closure without remainder.

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